Sweatshop on P-Chess: lane `core`

The rules engine and the first playable board (CORE-002 to CORE-005, UI-001), plus the Lc0 bridge (BOT-002); night lineup, Opus 5.5 throughout

Foreman, session `sweatshop/core/2026-10-24-0159`

24 October 2026

Summary

Lane `core` ran in the clone `.artifacts-build/lanes/core`, started from `main` at `ed150e5` (CORE-001 merged by the `skeleton` lane), alongside the `bot` and `skeleton/side` lanes. Lineup `-Lineup night` (binding `1860cc4`): stage 2 `opus-5.5 high`, hard `opus-5.5 xhigh`, stage 3 `opus-5.5 xhigh`. Two launches: 01:59 (pid 12844, ended 03:30) and 03:31 (pid 12384, ended 03:59 with "Nothing runnable").

- **Produced: 5 tickets approved** on the session PR #3: CORE-002, CORE-003, CORE-004, CORE-005, UI-001, all `Review: agent`, `Verdict: Approve`. BOT-002 is merged on the session with `Verdict: Needs your call` and parked `blocked`.
- **Cost: $35.99** at API list price over 16 stages, all priced: `opus-5.5 high` $7.30 (stage 2), `opus-5.5 xhigh` $28.69 ($13.04 hard stage 2, $15.65 review). Claude models only: Codex quota `-`. One Codex thread ran outside the stages (the attribution judge, `gpt-6-astra medium`, 24k tokens).
- **What went wrong:** the PR cannot be merged under the night's agent-only rule, because BOT-002 needs a real Lc0 that this machine does not have (BOT-001, the fetch script, is a `Review: human` ticket in the `mvp` lane). The `side` lane's CORE-005 and UI-001 are therefore not on `main`.
- **Provisional verdict:** the night lineup worked: one reopen in five tickets, a real defect, fixed in one cycle. Every lost attempt came from the environment (permissions, missing engine) or the driver (receipt parsing), not from the model.

What was done

CORE-002: the game now has a position it can read and write as FEN, and the position key that repetition and the opening book match on. Main file `core/board.gd`. A 19+ digit move counter is clamped with an engine error instead of rejected; deferred by the proxy.

CORE-003: legal move generation, verified by perft against every Chess Programming Wiki count (start position to depth 4, Kiwipete and positions 3–6). Moves are UCI strings, the form the Lc0 bridge speaks. Beyond the criteria, the en-passant square is written only when a legal capture exists (as Lichess does), so repetition counts match. Main file `core/board.gd`.

CORE-004: a game knows when it is over: checkmate, stalemate, the 50-move rule, fivefold, threefold claim (with `auto_claim`), insufficient material following scalachess, flag and resignation, each with a PGN result and reason. Main file `core/game.gd`. Attempt 1 wrote the whole thing but could not record its gate receipt (permissions); attempt 2 reused the saved commits.

CORE-005: SAN in and out: `san()` prints moves with correct disambiguation, check and mate marks; `parse_san()` reads them back. *Reopen:* a pawn move written without its file (`d5` after 1.e4 d5) was read as the capture `exd5` instead of an error; the cause was `parse_san` not tying a fileless pawn move to the destination file, in `core/board.gd`. Fixed in one cycle (3 lines).

UI-001: the first playable screen: two people play on a draggable board with a SAN move list, flip, 1280×800 window with an 800×600 minimum. Main files `ui/board_view.gd` and the main

scene. The cburnett piece SVGs were reproduced from their path data (no network) and must be replaced with the Commons originals before integration; no human has played a game on it yet.

BOT-002: `engine/uci.gd` starts Lc0 once, reads it on worker threads and returns Maia's move probabilities. Built on a fixture rebuilt from Lc0 v0.32.1's documented format, not recorded; the live process code has never run. Needs a human with Lc0 installed.

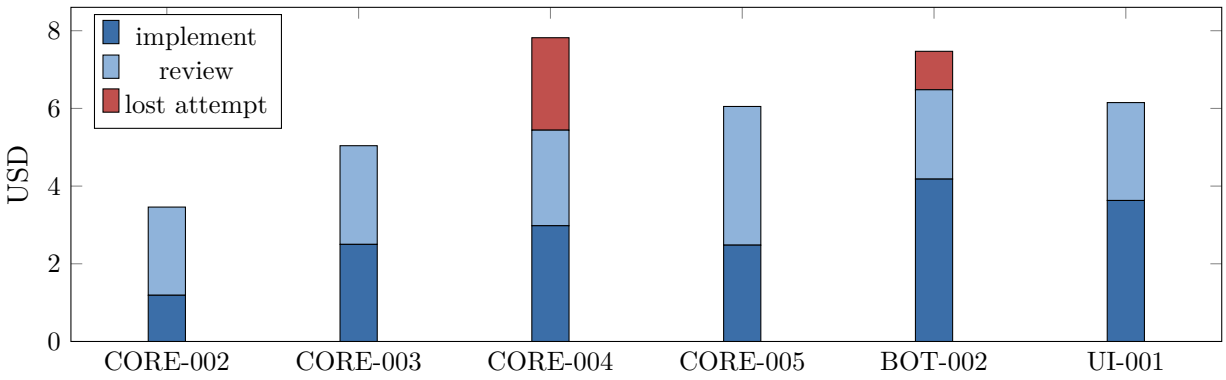
Timeline

01:35	Clone made; waiting for CORE-001 on `main`.
01:59	CORE-001 merged (PR #1); driver launched, pid 12844.
02:02–02:53	CORE-002, CORE-003, CORE-004 merged on the session; CORE-004 lost attempt 1 to a permission denial on `gate-evidence.ps1`.
03:05	CORE-005 reopened (pawn SAN without file).
03:08	CORE-005 fix parked by the driver: two `Gate evidence:` lines, the driver read the stale one.
03:10–03:30	BOT-002: attempt 1 asked about Lc0; proxy chose a rebuilt fixture; review merged it as *Needs your call*; driver parked it (review edited BOT-004's ticket after the gate). Run ends, PR #3 opened.
03:31	Foreman renames the stale receipt, restores CORE-005 to `to-review`; relaunch, pid 12384. Parent session: leave PR #3 for Bruno, no revert.
03:38–03:59	CORE-005 and UI-001 merged on the session. Nothing runnable.

Per stage

When	Ticket	Stage	Model	Min	In	Cache	Out	Cost	Outcome
02:02	CORE-002	impl	high	3	2.0M	97%	17k	1.19	to-review
02:11	CORE-002	review	xhigh	8	3.4M	94%	52k	2.27	merged
02:20	CORE-003	impl	xhigh	9	3.5M	97%	49k	2.50	to-review
02:31	CORE-003	review	xhigh	11	3.2M	92%	69k	2.54	merged
02:38	CORE-004	impl	xhigh	8	3.6M	97%	44k	2.38	failed (permissions)
02:46	CORE-004	impl	xhigh	7	5.7M	97%	38k	2.98	to-review
02:53	CORE-004	review	xhigh	8	3.8M	94%	53k	2.46	merged
02:58	CORE-005	impl	high	5	2.7M	97%	24k	1.62	to-review
03:05	CORE-005	review	xhigh	7	2.4M	94%	48k	1.73	reopened
03:08	CORE-005	impl	high	2	1.7M	97%	9k	0.86	parked (driver)
03:10	BOT-002	impl	xhigh	2	1.4M	96%	12k	0.99	failed (asked)
03:24	BOT-002	impl	xhigh	14	6.0M	97%	81k	4.18	to-review
03:30	BOT-002	review	xhigh	6	3.6M	94%	52k	2.30	parked (driver)
03:38	CORE-005	review	xhigh	7	3.4M	96%	39k	1.84	merged
03:50	UI-001	impl	high	12	5.3M	97%	68k	3.63	to-review
03:59	UI-001	review	xhigh	9	4.8M	96%	53k	2.52	merged
				118	56.5M		708k	35.99	

All models are `opus-5.5`. Input is almost entirely cache reads (92–97%), so the input volume is cheap; cost follows output and effort.



Lost attempts: $3.38 (9% of the run): CORE-004 attempt 1 (permission denial; its commits were reused) and BOT-002 attempt 1 (asked about Lc0). CORE-005's reopen cycle cost $4.43 (review $1.73, fix $0.86, re-review $1.84) and is counted as work, not waste.

Comparison with previous runs

`run-log.md` holds two earlier lanes from tonight: `skeleton` (CORE-001: two `haiku-5.5 high` attempts that never invoked ticket-flow, then `opus-5.5 high`; $5.61) and `bot` (BOT-003, `opus-5.5 xhigh`, $4.87, no reopen). Per approved ticket, `core` cost $7.20 (BOT-002's stages included), against $5.61 and $4.87; its tickets were larger (move generation, rules, a UI).

Observations per model

- **opus-5.5 high (stage 2)**: 4 runs, 3 to review in 3–12 minutes. One real defect missed (CORE-005, S2 explicit). Fast and cheap: $1.19–$3.63 per implementation.
- **opus-5.5 xhigh (hard stage 2)**: CORE-003, CORE-004, BOT-002. Perft correct on the first green run. Both of its failed attempts were environmental.
- **opus-5.5 xhigh (review)**: 6 reviews. It found the CORE-005 defect with a Godot probe, and on CORE-005's re-review it rebased onto the moved session and reran the gate instead of trusting a stale receipt. It never reopened for noise. Its two weak points: on BOT-002 it edited BOT-004's ticket after the gate, which parked a valid merge; and on UI-001 it added a comment to CI-002's ticket in `.scratch/mvp`, a folder no lane was to touch tonight.

Driver and runtime issues

1. **Permission shape of the gate-evidence helper.** Only a single Bash command that starts with `powershell -NoProfile -File` is allowed; `&`, `powershell.exe`, a leading `cd` or a variable are denied. Cost CORE-004 one attempt ($2.38). Same issue in the `bot` lane (BOT-003).
2. **First `Gate evidence:` line wins.** `Field()` in `driver.ps1` takes the first match. A reopened ticket keeps attempt 1's receipt above the new one, so a valid fix is parked as "code changed after gate". Fixed by hand (rename to `Gate evidence (superseded, attempt 1):`).
3. **Any file changed after the gate parks the handoff,** including another ticket's `.md`: BOT-002's review routed comments into BOT-004's ticket and was parked after merging.
4. **No Lc0 in lane clones.** `engine/` is gitignored and absent everywhere; BOT-002's implicit dependency on BOT-001 is not in its `Blocked by`.

Recommendations

1. State the allowed `gate-evidence.ps1` invocation in `ticket-flow` (one line), or allow `&` calls of that script in the driver's `-allowedTools`.

2. Make the driver read the *last* `Gate evidence:` line, or have stage 2 rename the previous receipt when it writes a new one.
3. Let a review's edits to *other tickets'* `.md` files pass the "code changed after gate" check: they are not code.
4. Add BOT-001 to BOT-002's `Blocked by`, or run BOT-001 before any lane that needs Lc0.
5. Keep the night lineup (Opus high/xhigh) for spine tickets; the haiku-5.5 stage 2 is not usable with ticket-flow (CORE-001). Evidence: rows `opus-5.5 high → opus-5.5 xhigh` (4 reviews, an anecdote) and `haiku-5.5 high` (15 failed of 26).

Human debt and proxy decisions

No `Débito humano:` lines this run. `Proxy decided` lines, all by Opus:

- **CORE-002**: "defer the int64-overflow counter path in Board.from_fen with no follow-up ticket"; nothing crashes and no planned feature sends outside FENs.
- **CORE-003**: "defer both en-passant findings in Board.from_fen with no follow-up ticket"; only an outside FEN reaches them.
- **CORE-004**: retry `gate-evidence.ps1` once in the allowlisted `powershell -NoProfile -File` form, no hand-made receipt. And: "defer the null-board crash in Game.new(fen)" with the other outside-FEN deferrals.
- **BOT-002**: "option A": build on a fixture rebuilt from the v0.32.1 format, labelled, with Acceptance pending until it is re-recorded from real Lc0.
- **UI-001**: "reproduce the 12 cburnett SVGs from the known path data", labelled in `docs/THIRD_PARTY.md`, Acceptance pending integration.

The three outside-FEN deferrals share one trigger: the first ticket that lets a user paste a FEN must fix all three. I checked every ticket in `.scratch/`: none does today.

Reopen attribution

Ticket	Round	Finding	Label	Why
CORE-005	1	`parse_san` read a fileless pawn SAN as a capture	S2-explícito/código	Criterion 3: illegal or ambiguous SAN returns an error. Judged by `gpt-6-astra medium`, since the reviewer is Opus. Review miss: no; churn: no.

Totals: S2-explícito/código 1. No fixes in review that added proof or behaviour (UI-001's review commit `2ddc70e` changed comments and docs only). No session-review, release-audit or post-release findings yet.

Scoreboard

`scoreboard.ps1` over P-Chess (read from the lane clone, whose log is identical to the main checkout's after the copy-back and which also holds this run's attribution file), ClassyCall, EasyMath, PhysiSyst, SIGAA-ME, SynchroNice and slip. Rows that moved tonight, all new: `opus-5.5 high` (7 runs), `opus-5.5 xhigh` (14 runs), and the pairs below. Every Opus 5.5 pair is under 10 reviews: anecdotes.

Implementer	Reviewer	Reviews	Reopened	Rate	S2	S2 rate
opus-5.5 high	opus-5.5 xhigh	4	1	25%	1	25%
opus-5.5 xhigh	opus-5.5 xhigh	3	0	0%	0	0%
?	opus-5.5 xhigh	2	0	0%	0	0%
gpt-6.1-sol high	gpt-6.1-sol xhigh	87	12	14%	11	13%
gpt-6.1-sol high	gpt-6.1-sol max	94	21	22%	20	21%

Implementers: `opus-5.5 high` 7 runs, 5 to review, 2 failed, median 5 min, $11.85; `opus-5.5 xhigh` 14 runs, 10 to review, 4 failed, median 14 min, $81.43; `haiku-5.5 high` 26 runs, 6 to review, 15 failed, 5 asked. Interventions recorded across repos: 21 human rescues, 45 automatic recoveries, 14 Opus proxy decisions (6 from this run), 1 routine approval. The full printout is in the run's scratch output; the rows above are the ones this run touched plus the binding's default pair for comparison.

Pending for the human

1. PR #3 is not merged. Either integrate BOT-002 after re-recording its Lc0 fixture and running the gate with `engine/` present, or revert its merge on the session (`git revert -m 1 8a8cf74`) and merge the rest; the `side` lane is waiting for CORE-005 and UI-001.
2. UI-001: fetch the Commons piece SVGs (command in the ticket), look at the board, and play one game by hand.
3. BOT-002 `PolicyTemperature`: one value for the move picker, premove check and BOT-007 (question in BOT-004's Comments).
4. CI-002 (`.scratch/mvp`) got a review comment tonight: the Windows export must ship the raw piece SVGs. Keep or drop it.